LUND UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Number Theory MATC35
18th of December 2025

Suggested Solutions

1. **(1+2+2 points)** Determine whether the following congruences equations are solvable, and solve them completely in case they are:

(a) $x^2 + x - 2 \equiv 0 \ (\mod 17)$.

(b) $3y^4 - 4y^2 + 2 \equiv 0 \ (\mod 11^2)$.

(c) $3x^{15} \equiv 2 \ (\mod 13)$. (Hint: You may use that 2 is a primitive root of 13).

Solution:
(a.) Multiplying by 4 which is relatively prime to 17, and completing the square, we obtain that the equation is equivalent to $(2x+1)^2 \equiv 9 \ (\mod 17)$. A simple inspection readily shows that $x = 1, 16$ are the two incongruent solutions modulo 17.
(b.) Multiplying the equation with 3, being relatively prime to 11, and completing the square in y^2, we see that for the equation is solvable if and only if

$$(3y^2 - 2)^2 \equiv -5 \ \left(\mod 11^2\right).$$

Therefore, for the equation to admit a solution it is necessary that $(-5/11) = 1$. Using the Gauss law of quadratic reciprocity and rules for the Legendre symbol, we get

$$(-5/11) = (-1/11)(5/11) = -(11/5) = -(1/5) = -1$$

hence the equation has no solution.
(c.) Since 2 is a primitive root of the prime 13, we may use the theory of indices and transform the polynomial equation to:

$$\mathrm{ind}_2(3) + 15\mathrm{ind}_2(x) \equiv \mathrm{ind}_2(2) \ (\mod 13).$$

Making a list of the congruences of $2^k \equiv$ modulo 13 for $k = 1, 2, \ldots, 12$, we verify that $\mathrm{ind}_2(3) = 4$ and $\mathrm{ind}_2(2) = 1$. Setting $y = \mathrm{ind}_2(x)$ the equation reduces to

$$4 + 15y \equiv 1 \ (\mod 12) \iff 12|15y + 3 = 3(5y + 1) \iff 4|y + 1.$$

However, there are precisely 3 such incongruent solutions y modulo 12, namely $y = 3, 7, 11$. These give rise to the 3 incongruent solutions $x_1 = 2^3 = 8$, $x_2 = 2^7 \equiv 11$, $x_3 = 2^{11} \equiv 7$ modulo 13 to our initial polynomial equation.

3. **(2 + 3 points)** Let a, b be integers both relatively prime to $n > 1$, and consider the orders of a, b modulo n denoted by $h = \mathrm{ord}_n(a)$ and $k = \mathrm{ord}_n(b)$.

(a.) Show that the order of ab modulo n divides hk.
(b.) Show that if $\gcd(h,k)=1$, then $\mathrm{ord}_n(ab)=hk$.
(Hint: If $d=\mathrm{ord}_n(ab)$ then part **(a.)** implies $hk=dm$ for some $m\geq 1$. If $m>1$ then it has a prime factor p, which must either divide h or k, but not both!).
Solution:

(a.) Note that

$$(ab)^{hk}=(a^h)^k\cdot(b^k)^h\equiv 1^k\cdot 1^h=1 \pmod n,$$

hence $\mathrm{ord}_n(ab)|hk$ by Theorem from the book.
(b.) Let $d=\mathrm{ord}_n(ab)$ and note that from **(a.)**, we have that $hk=dm$ for some integer $m\geq 1$. If $m>1$, then we can find a prime factor p of m, which has the property that $p|hk/d$. This implies that $hk=dpl$ for some integer $l\geq 1$. Since p is a prime, we must have either h or k is divisible by p. We without loss of generality assume that $p|h$. Then we have

$$1\equiv(ab)^{dl}=(ab)^{hk/p}=a^{hk/p}(b^k)^{h/p}\equiv a^{hk/p} \pmod n$$

which implies that $h=\mathrm{ord}_n(a)|hk/p$, that is, $p|k$. But this implies that $p|\gcd(h,k)=1$, which is a contradiction. We conclude that $m=1$, hence $d=hk$.

4. **(5 points)** Let p be a prime of the form $p\equiv 3 \pmod 4$. Show that the product of all the quadratic residues of p is congruent to 1 modulo p.
Solution: Let r be a primitive root of p and recall that the quadratic residues of p are precisely the numbers of the form $r^2, r^4, \ldots r^{2(p-1)/2}$ by Euler's criterion, and they are $(p-1)/2$ incongruent numbers modulo p. This implies that

$$\prod_{k=1}^{(p-1)/2} r^{2k}=r^{\sum_{k=1}^{(p-1)/2}2k}=r^{\frac{(p-1)}{2}\frac{(p+1)}{2}}\equiv(-1)^{\frac{(p+1)}{2}} \pmod p.$$

Since p is an odd prime, we see that the later expression is 1 if (and only if) $p\equiv 3 \pmod 4$.

5. **(2+3 points)** A positive integer $n>1$ is said to be a *perfect number* if $\sigma(n)=2n$, where σ denotes sum of divisor function.
(a.) Show that if 2^k-1 is a prime number for some $k>1$, then $n=2^{k-1}(2^k-1)$ is a perfect number.
(b.) Show that every even perfect number n is of this form.
(Hint: Write $n=2^km$ with $k\geq 1$, m odd and consider the expression $2n=\sigma(n)$).
Solution:

(a.) Assume that $p=2^k-1$ is a prime. Then if $n=2^{k-1}(2^k-1)$, we have by the multiplicative property of σ that

$$\sigma(n)=\sigma(2^{k-1})\sigma(2^k-1)=(1+2+\cdots+2^{k-1})\cdot 2^k=2\cdot 2^{k-1}(2^k-1)=2n.$$

(b.) Assume n is an even perfect number, and note that the fundamental theorem of arithmetic allows us to write $n=2^km$, where $k\geq 1$ and m is odd. The assumptions imply that

$$2^{k+1}m=2n=\sigma(n)=\sigma(2^k)\sigma(m)=(2^{k+1}-1)\sigma(m).$$

Since $\gcd(2^{k+1}, 2^{k+1}-1) = 1$, we must have that $2^{k+1}-1$ divides m: $m = (2^{k+1}-1)l$ for some $l \geq 1$. Returning to the previous equation, we get

$$2^{k+1}l = \sigma((2^{k+1}-1)l)$$

Note that if $l > 1$, then by the definition of the divisor function, we have

$$2^{k+1}l = \sigma((2^{k+1}-1)l) \geq (2^{k+1}-1)l + l + 1 = 2^{k+1}l + 1,$$

which is absurd, hence $l = 1$. This implies that $n = 2^k(2^k - 1)$ and we must only show that $2^k - 1$ is a prime. But this readily follows from the condition

$$2^{k+1} = \sigma(2^{k+1} - 1),$$

hence the proof is complete.

6. (5 points) Prove that an odd prime number p can be expressed of the form

$$p = x^2 + 3y^2$$

for some integers x, y if and only if $p \equiv 1 \pmod 6$.
Solution:

Assume that $p = x^2 + 3y^2$ for some integers $x, y \in \mathbb{Z}$. Since $\gcd(y, p) = 1$, there exists an integer z such that $yz \equiv 1 \pmod p$. Then

$$(xz)^2 = x^2z^2 \equiv -3y^2z^2 = -3(yz)^2 \equiv -3 \pmod p.$$

This is equivalent to $(-3/p) = 1$ which is equivalent to $p \equiv 1 \pmod 6$ (for instance, this can proved from Gauss's law of reciprocity). Conversely, if $p \equiv 1 \pmod 6$ then $(-3/p) = 1$ and hence there exists a number a such that

$$a^2 \equiv -3 \pmod p.$$

Since $p > 3$ we must have $\gcd(a, p) = 1$, and appealing to Thue's Lemma, we can find integers x, y with $0 < |x|, |y| < \sqrt{p}$, such that $ay \equiv x \pmod p$. This implies that

$$x^2 \equiv (ay)^2 = a^2y^2 \equiv -3y^2 \pmod p.$$

This implies that $pk = x^2 + 3y^2$, and since $0 < |x|, |y| < \sqrt{p}$, we get $k = 1, 2, 3$. If $k = 1$ then p is clearly of the desired form. If $k = 2$, we see that $2p = x^2 + 3y^2$ implies that either both x, y are even or x, y are odd. In any case, we get

$$2p = x^2 + 3y^2 \equiv 0 \pmod 4$$

which absurd, hence $k \neq 2$. If $k = 3$, then we can infer that $3|x$. Writing $x = 3x_1$, we get

$$3p = 3^2x_1^2 + 3y^2 \implies p = 3x_1^2 + y^2,$$

and p is of the same desired form. We have therefore produced an expression of the form $p = x^2 + 3y^2$, whenever $p \equiv 1 \pmod 6$.